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Inventor(s) Chen; Tong et al.

Self-mixing interference device with tunable microelectromechanical system

Abstract

Self-mixing interferometry (SMI) sensors may include vertical cavity surface emitting lasers (VCSEL), photodetectors, and microelectromechanical systems (MEMS). The VCSEL, photodetectors, and MEMS may be vertically stacked. The MEMS may be moveable with respect to a VCSEL and may change a cavity length associated with the VCSEL. By changing the cavity length associated with the VCSEL, certain properties of emitted light may be changed, such as a wavelength value of the emitted light.

Inventors: **Chen; Tong (Fremont, CA), Cihan; Ahmet Fatih (San Jose, CA), Vail; Edward (Menlo Park, CA), Li; Weiping (Pleasanton, CA), Fang; Xiaolong (Fremont, CA), Zhou; Xibin (Palo Alto, CA), Qiao; Pengfei (El Cerrito, CA)**

Applicant: **Apple Inc.** (Cupertino, CA)

Family ID: **85572867**

Assignee: **Apple Inc. (Cupertino, CA)**

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Primary Examiner: Park; Kinam

Attorney, Agent or Firm: Brownstein Hyatt Farber Schreck, LLP

Background/Summary

FIELD

(1) Embodiments described herein generally relate to structures and configurations of laser diodes, such as vertical cavity surface emitting lasers (VCSELs), and associated photodetectors, such as resonant cavity photodetectors (RCPDs). Laser diodes and associated photodetectors may be parts of detector systems that measure or determine displacement, distance, motion, speed, or velocity of a surface or an object using self-mixing interferometry (SMI).

BACKGROUND

(2) Laser diodes are used for a number of purposes, such as material processing, spectroscopy, and information processing. For some applications, a wavelength of an emitted laser beam may need to be periodically changed or tuned. Tuning may be performed by changing a temperature of the laser diode. However, such methods of tuning can be time-consuming and may offer a narrow tuning range. In addition, changing a temperature of a laser diode may degrade the laser diode and/or result in a shortened lifespan of the laser diode.

SUMMARY

(3) This summary is provided to introduce a selection of concepts in simplified form that are further described herein. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(4) In some embodiments, a self-mixing interferometry (SMI) sensor may be provided. A SMI sensor may comprise a photodetector comprising a first set of semiconductor layers formed on a substrate, a vertical cavity surface emitting laser (VCSEL) stacked with the photodetector and comprising a second set of semiconductor layers formed on the first set of semiconductor layers, and a microelectromechanical system (MEMS) stacked with the VCSEL on a side opposite from the photodetector and comprising an at least partially reflective layer. The VCSEL may be configured to emit light and the MEMS may be configured to change a length of a resonant cavity associated with the VCSEL, thereby changing a property of the emitted light.

(5) The VCSEL may be further configured to receive reflections of the emitted light after the emitted light interacts with an object and undergo self-mixing interference after receiving the reflections of the emitted light.

(6) A SMI sensor may further comprise a processor. At least a portion of the emitted light may be received by the photodetector and, based on at least the portion of the emitted light received by the photodetector, the processor may determine at least one of a speed of the object, a roughness of the object, or a distance between the SMI sensor and the object.

(7) The MEMS may be separated from the VCSEL by a void in a semiconductor layer.

(8) An SMI sensor may further comprise processing circuitry configured to supply a first current to the VCSEL, the first current configured to drive the VCSEL and supply a second current to the MEMS, the second current separate from the first current and configured to cause a movement of the MEMS. A changed property of the emitted light may be a wavelength of the emitted light.

(9) An at least partially reflective layer of the MEMS may be formed from at least one of a third set of semiconductor layers or a set of dielectric layers.

(10) According to some embodiments, a tunable sensor for performing self-mixing interferometry may be provided. The tunable sensor may comprise a vertical cavity surface emitting laser (VCSEL) comprising a laser cavity, a microelectromechanical system (MEMS) comprising an at least partially reflective layer defining an end of the laser cavity, and a photodetector positioned to receive light emitted by the VCSEL after the light passes through the at least partially reflective layer. A position of the at least partially reflective layer, which may be moved when a voltage is

applied to the MEMS, may define a length of the laser cavity.
(11) The VCSEL may be a first VCSEL provided on a first set of semiconductor layers, a second VCSEL may be provided on the first set of semiconductor layers, the second VCSEL having a second laser cavity and the first VCSEL and the second VCSEL being separated by a trench, and the at least partially reflective layer of the MEMS may extend across the first VCSEL and the second VCSEL and may further define a second length of the second laser cavity.
(12) The MEMS may change the length of the laser cavity by moving the at least partially reflective layer from a first position to a second position. Changing the length of the laser cavity may result in changing a property of the light emitted by the VCSEL. The changed property of the light emitted by the VCSEL may be a wavelength of the light.
(13) The tunable sensor may further comprise a condenser lens disposed in a light emission path of the VCSEL on a first surface opposite from a second surface where the MEMS is coupled to the VCSEL.
(14) The VCSEL may comprise an active layer, an oxide layer adjacent to the active layer, a first set of distributed Bragg reflectors positioned between the MEMS and the oxide layer, and a second set of distributed Bragg reflectors positioned between the condenser lens and the active layer.
(15) The photodetector may be spaced apart from the MEMS and the VCSEL.
(16) In some embodiments, a self-mixing sensor may be provided. The self-mixing sensor may comprise a substrate, a first set of semiconductor layers positioned above the substrate, the first set of semiconductor layers operating as a first light receiving or light emitting element, a second set of semiconductor layers stacked with the first set of semiconductor layers, the second set of semiconductor layers operating as a second light receiving or light emitting element, and a microelectromechanical system (MEMS) stacked with the second set of semiconductor layers, the MEMS comprising a moveable reflective layer.
(17) The self-mixing sensor may further comprise processing electronics electrically coupled with the first set of semiconductor layers, the processing electronics configured to analyze the electrical current produced by the first set of semiconductor layers to determine self-mixing properties of the light.
(18) The first set of semiconductor layers and the second set of semiconductor layers may each comprise distributed Bragg reflectors.
(19) The moveable reflective layer may comprise at least one of a diffractive grating or a subwavelength grating to polarize the light produced by the VCSEL.
(20) The self-mixing sensor may further comprise a void positioned between the MEMS and the second set of semiconductor layers, the void and the second set of semiconductor layers at least partially defining a laser cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Reference will now be made to representative embodiments illustrated in the accompanying figures. It should be understood that the following descriptions are not intended to limit the embodiments to one or more preferred embodiments. To the contrary, they are intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the described embodiments as defined by the appended claims. Similar reference numerals have been used, where practicable, to designate similar features.
(2) FIGS. **1**A-**1**B illustrate an example configuration of a self-mixing sensor comprising a vertical cavity surface emitting laser (VCSEL), a vertically integrated photodetector, and a vertically integrated microelectromechanical system (MEMS) for detecting properties of an object and/or a surface, as discussed herein.

(3) FIG. **2**A depicts an example self-mixing interference process occurring with respect to a self-mixing sensor, as discussed herein.
(4) FIG. **2**B depicts a graph showing a variation in a power level of laser light emitted from a VCSEL undergoing self-mixing interference with respect to length of a feedback cavity, as discussed herein.
(5) FIG. **2**C depicts time correlated graphs of a laser current, laser wavelength, and a signal of an interferometric parameter that can be used as part of a spectrum analysis, as discussed herein.
(6) FIG. **3** illustrates an example configuration of a self-mixing sensor comprising a VCSEL-integrated photodetector (ViP) and an associated MEMS, as discussed herein.
(7) FIG. **4** illustrates example processing circuitry communicatively coupled with electrical contacts of the self-mixing sensor of FIG. **3**, as discussed herein.
(8) FIG. **5**A illustrates an example inverted self-mixing sensor formed from a MEMS stacked with a VCSEL, as discussed herein.
(9) FIG. **5**B illustrates an example inverted self-mixing sensor including a MEMS stacked with a photodetector, as discussed herein.
(10) FIG. **5**C illustrates an example inverted self-mixing sensor including a surface grating, as discussed herein.
(11) FIG. **6** illustrates an example high-powered self-mixing sensor formed with an expanded-cavity ViP, as discussed herein.
(12) FIG. **7**A illustrates an example self-mixing sensor array, as discussed herein.
(13) FIG. **7**B illustrates an example self-mixing sensor array including a surface grating, as discussed herein.
(14) FIG. **8** illustrates an example self-mixing sensor where a photodetector is positioned below a VCSEL, as discussed herein.
(15) FIG. **9** illustrates an example self-mixing sensor where a photodetector is positioned on a top portion of the self-mixing sensor, as discussed herein.
(16) FIG. **10** depicts an example method for manufacturing a self-mixing sensor, as discussed herein.
(17) The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures. Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.
(18) Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented therebetween, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

(19) The embodiments described herein are directed to self-mixing interferometry (SMI) sensors. Described embodiments may include one or more SMI sensor and one or more microelectromechanical systems (MEMSs) for varying a cavity length of the SMI sensor.
(20) A SMI sensor is defined herein as a sensor configured to generate electromagnetic radiation (e.g., light), emit the electromagnetic radiation from a resonant cavity (e.g., a resonant optical cavity), receive a returned portion of the electromagnetic radiation back into the resonant cavity, coherently or partially coherently self-mix the generated and returned electromagnetic radiation within the resonant cavity, and produce an output indicative of self-mixing (e.g., a SMI signal). The

generated, emitted, and returned electromagnetic radiation may be coherent of partially coherent. In some examples, the electromagnetic radiation emitted by a SMI sensor may be generated by an electromagnetic radiation source such as a laser, which may be a vertical cavity surface emitting laser (VCSEL), a vertical external cavity surface emitting laser (VECSEL), a quantum dot laser (QDL), a quantum cascade laser (QCL), an edge emitting laser (EEL), a horizontal cavity surface emitting laser (HCSEL), a solid state laser (SSL), a gas laser, and so on. In additional or alternative examples, an electromagnetic radiation source may be a light emitting diode (LED), such as an organic LED (OLED), a resonant cavity LED (RC-LED), a micro LED (mLED), a superluminescent LED (SLED), an edge emitting LED, and so on. The generated, emitted, and returned electromagnetic radiation may include, for example, visible or invisible light, such as green light, red light, infrared (IR) light, ultraviolet (UV) light, and so on. The output of a SMI sensor (e.g., the SMI signal) may include a photocurrent produced by a photodetector, such as a photodiode, which is integrated with, or positioned under, above, or next to, the sensor's electromagnetic radiation source. Alternatively or additionally, the output of a SMI sensor may include a measurement of the current or junction voltage of the SMI sensor's electromagnetic radiation source.

(21) According to some aspects of the present disclosure, vertical cavity surface emitting lasers (VCSELs) and associated photodetectors, such as resonance cavity photodetectors (RCPDs), are provided as a SMI sensor, though any other type of SMI sensor may be used in accordance with the provided disclosure. A VCSEL-integrated photodetector (ViP) may be provided and may reference an arrangement of a VCSEL and photodetector formed within the same epitaxial stack. Additionally or alternatively, a VCSEL and photodetector may be provided as a stacked VCSEL and photodetector. A VCSEL and photodetector may be used as a touch or an input sensor, such as may be used within an electronic device such as an electronic watch, wearable electronics, a smart phone, and so on. One or more VCSELs and/or photodetectors may perform self-mixing interference as part of a self-mixing sensor and may be used to detect a presence, proximity, displacement, distance, motion, speed, roughness, or velocity of an input surface, such as a touch screen, a button, one or more particles, or a body part. Hereinafter, for convenience, all such possibly measured parameters will be referred to simply as “displacement or motion.”

(22) Self-mixing sensors may detect the displacement or motion of an object by causing one or more VCSELs to emit a laser light toward an object by applying a forward bias to the VCSEL. Reflections of the emitted laser light from the object can be received back into the lasing cavity of the VCSEL diode. This can cause self-mixing interference in which there is a shift in the phase of a laser mode and of the emitted laser light to a new altered value.

(23) A power shift may be concurrently detected by a photodetector which in turn can be related to the phase shift induced by, for example, object surface and volume displacement or object distance in wavelength-swept laser operations. Processing circuitry (e.g., a processor or other circuit) included in the self-mixing sensor may then be able to determine the spatial property of objects through temporal or spectral analysis of a photodetector signal. For example, a VCSEL and photodetector may be used as a self-mixing interferometry (SMI) sensor according to the above discussed principles and a microelectromechanical system (MEMS) may be affixed to a top surface of the VCSEL or the photodetector.

(24) A SMI sensor may be used for monolithic coherent optical sensing, where the SMI sensor obtains spatial information of a target in, for example, a heterodyne detection scheme. Wide-range and fast wavelength modulation of the VCSEL may be used in order to obtain a high spatial resolution and sensing bandwidth, for a variety of detected objects and applications. Typically, VCSEL wavelength modulation can be performed by modulating an input bias current, thereby causing a change in cavity temperature and effective cavity length of the VCSEL. However, modulating the VCSEL input bias current has several potential limitations, such as a low wavelength tuning range, a slow tuning speed, temperature-induced power and wavelength non-

linearity, and so on.

(25) Aspects of the present disclosure discuss a microelectromechanical system (MEMS) structure affixed to, or otherwise in proximity to, a VCSEL structure. The MEMS may include a tunable reflective, or partially reflective, layer which may be electrostatically, electrothermally, electromagnetically, and/or piezoelectrically actuated independently of the VCSEL. In this way, the MEMS may move independently, or partially independently, from the VCSEL thereby changing a length of an internal cavity, and therefore an emitted wavelength, of the VCSEL in a controllable manner.

(26) Movement of the MEMS may change an associated VCSEL cavity length by hundreds of nanometers, which can result in tuning ranges of approximately 0.1%-1%, 1%-5%, or greater than 5% of the wavelength emitted by the VCSEL, though these values may be any values in accordance with a free-spectral range and gain bandwidth of the VCSEL. Further, a modulation speed of the wavelength tuning may be on the megahertz order of magnitude, resulting in a tuning speed faster than cavity-thermal-based tuning speeds. Further, a tuning range may be larger than cavity-thermal-based tuning ranges. Additionally or alternatively, the VCSEL and/or photodetector structures may maintain relatively constant power and junction temperatures over a wavelength tuning range as the MEMS moves, resulting in linear tuning. In some cases, MEMS-based tuning may be used in addition to modulating a VCSEL bias current, to provide additional flexibility in tuning a VCSEL. As used herein, the term “tuning,” and any variation thereof, refers to changing a property of a VCSEL, such as changing a wavelength emitted by a VCSEL.

(27) Various embodiments described below describe structures or configurations of the one or more VCSELs, photodetectors, and MEMS structures that may be included as parts of self-mixing sensors. A VCSEL may be stacked on a RCPD, and a MEMS structure may be stacked on top of the VCSEL, by epitaxially forming semiconductor layers and etching portions of the semiconductor layers. Alternative methods of forming the common set of semiconductor layers may be used in addition to or instead of epitaxial growth. For simplicity of explanation, the deposition or growth of the common set of semiconductor layers will be described as formed epitaxially, but one skilled in the art will recognize that alternative methods are within the scope of this disclosure. In some embodiments, a SMI structure may be flipped, so that a VCSEL is stacked, or otherwise positioned, on top of a MEMS structure and/or a photodetector structure.

(28) These and other embodiments are described below with reference to FIGS. **1**A-**8**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these drawings is for explanatory purposes only and should not be construed as limiting.

(29) FIG. **1**A illustrates a self-mixing interferometry configuration **100** for a self-mixing sensor **102**. As depicted in FIG. **1**A, the self-mixing sensor **102** includes a VCSEL **104** (e.g., a VCSEL diode), a photodetector **106** (e.g., a resonant cavity photodetector (RCPD)) vertically stacked with the VCSEL **104**, and a microelectromechanical system (MEMS) **108** vertically stacked with the VCSEL **104** and the photodetector **106**. The VCSEL **104** may be configured to emit coherent or partially coherent laser light **112**, which may be directed toward an object **116**. The object **116** may be moveable along a three-dimensional space with respect to the self-mixing sensor **102** and may move closer to the self-mixing sensor **102**, further away from the self-mixing sensor **102**, and/or along a horizontal plane with respect to the self-mixing sensor **102**. Although the object **116** is depicted as a relatively planar object, the is only one particular example. An object may be a user's skin, a touch surface of a device, a watch crown shaft surface, one or more particles, and so on.

(30) As the laser light **112** contacts the object **116**, reflections **114** of the laser light **112**, otherwise referenced as backscatters, may reflect off of the object **116** in multiple directions. Some of the reflections **114** may be received back into a lasing cavity of the VCSEL **104**, causing the VCSEL **104** to undergo self-mixing interference and thereby altering a property (e.g., a wavelength or power) of the emitted laser light **112**. In additional or alternative cases, an electrical property of the VCSEL **104** is altered during self-mixing, such as a junction voltage that varies corresponding to

with a distance or motion of the object **116** with respect to the self-mixing sensor **102**.
(31) The photodetector **106** may detect an operational change (e.g., a power change) in the VCSEL **104** due to the self-mixing interference. For example, the VCSEL **104** may emit some laser light downwards into the photodetector **106**. Any alterations in the emitted laser light **112** due to self-mixing interference may also occur in the downward emitted laser light, and cause an operational change in the photodetector **106**. For example, a variation in a photocurrent I.sub.PD of the photodetector **106** may be detected by a current monitor **118**. As depicted in FIG. **1**A, a photocurrent I.sub.PD may flow out of a back side of the photodetector **106** to establish a reverse voltage bias.
(32) The MEMS **108** may be positioned on a top surface of the VCSEL **104**. In some cases, the MEMS **108** includes a reflector, such as a surface grating (e.g., a diffractive or subwavelength grating), a meta-structure, a stack of thin-film reflectors, or a Distributed Bragg Reflector (DBR), that expands an internal lasing cavity of the VCSEL **104**. A voltage V.sub.MEMS may additionally be applied to the MEMS **108**. When the voltage V.sub.MEMS is applied, the MEMS **108** may move or may deform with respect to a top surface of the VCSEL **104**. As the MEMS **108** moves or deforms, an internal lasing cavity of the VCSEL **104** changes, thereby changing a wavelength of emitted laser light. The movement or deformation of the MEMS **108** may be controllable through control of a value of the V.sub.MEMS and/or a duration when the V.sub.MEMS is applied. As described herein, the MEMS may be electrostatically actuated. In some implementations, the MEMS may be actuated electrothermally, electromagnetically, or piezoelectrically.
(33) The VCSEL **104** may emit the laser light **112** under a forward voltage bias above a laser threshold. During such a forward voltage bias, a bias current I.sub.BIAS **110** flows through the VCSEL **104** and may induce laser light emission from the VCSEL **104**. The photodetector **106** may be reverse biased during at least part of the time that the VCSEL **104** is forward biased.
(34) FIG. **1**B illustrates an exemplary configuration of a self-mixing sensor **102**, such as described above with respect to FIG. **1**A. The self-mixing sensor **102**, as depicted, includes a VCSEL **104**, a photodetector **106**, and a MEMS **108**. The VCSEL **104**, as discussed above, may be part of the self-mixing sensor **102** and, in some cases, other types of electromagnetic radiation sources may be used. In any type of laser, an input energy source causes a gain material within a cavity to emit light. Reflectors on opposite ends of the cavity feed the light back into the gain material to cause amplification of the light and to cause the light to become coherent, with a uniform wavelength, or partially coherent, with a partially uniform wavelength. At least one reflector may be partially reflective and may permit light to escape a cavity under certain conditions (e.g., a strength/coherence of light within a laser cavity).
(35) In the VCSEL **104**, two reflectors may be provided on opposite ends of the laser cavity, where one reflector may be defined by the MEMS **108**. The lasing process occurs within the defined laser cavity. In the case of VCSELs, the two reflectors may be implemented as distributed Bragg reflectors, which are alternating reflective layers with high and low refractive indices. The laser cavity contains a gain material, which may include multiple doped layers of III-V semiconductors. Specific details of the semiconductor materials will be presented below for the various embodiments. The emitted laser light can be emitted through the topmost layer or surface of the VCSEL **104**. In the VCSEL **104**, the coherent light is also emitted through the bottom layer and is received by the photodetector **106** so that self-mixing operations may be analyzed and a motion or distance of an object upon which the VCSEL **104** emits light can be determined.
(36) As discussed with respect to FIG. **1**A, a MEMS **108** may be provided at a top portion of the VCSEL **104**. The MEMS **108** may be moveable and may act as a partially transmissive reflector defining one end of a laser cavity. Accordingly, the MEMS **108** may be formed from one or more distributed Bragg reflectors (DBR). As the MEMS **108** moves relative to a top surface of the VCSEL **104**, a length of the laser cavity of the VCSEL **104** changes, resulting in a wavelength of emitted light **112** changing. The MEMS **108** may be coupled to the VCSEL **104** through a

sacrificial DBR layer (e.g., a DBR layer only provided for structural purposes and not for a potential separate utility as a reflective layer) and may be provided in a cantilevered state, though any MEMS **108** structure capable of expanding or contracting a VCSEL **104** laser cavity may be used in accordance with the provided disclosure. In some cases, the MEMS **108** is supported on one or more sides with respect to the VCSEL **104**.

(37) FIGS. **2**A-**2**C illustrate properties of self-mixing interference of emitted laser light in a VCSEL. The explanations are intended only to describe certain aspects of self-mixing interference needed to understand the disclosed embodiments. Other aspects of self-mixing interference will be clear to one skilled in the art.

(38) FIG. **2**A depicts a functional diagram of a laser device, such as a VCSEL described with respect to FIGS. **1**A-**1**B, that may be used as part of a self-mixing sensor. As discussed above, in any type of laser, an input energy source causes a gain material within a cavity to generate light. Reflectors **202** and **204***a* on opposite ends of a laser cavity **206** feed the light back into the gain material to cause amplification of the light within the laser cavity **206** and to cause the light to become coherent, with a uniform wavelength, or partially coherent, with a partially uniform wavelength. In some implementations, one of reflectors **202**/**204***a* may be partially reflective and may permit light to pass through under certain conditions (e.g., a coherence/intensity of light within the laser cavity **206**). As discussed below, the reflector **204***a* may be moveable by a distance **205**, as indicated by reflector **204***b*, depicted in phantom. That is, during a tuning process, the reflector may change positions, thereby changing a length of the laser cavity **206**.

(39) In FIG. **2**A, the laser cavity **206** is shown reoriented so that the emitted laser light is emitted from the laser cavity **206** to the right. The laser cavity **206** has a length established at manufacture and the length may be tuned, or changed, due to movement of an associated MEMS (e.g., a reflector that moves between **204***a* and **204***b* as depicted in FIG. **2**A). The emitted laser light travels away from the laser cavity **206** until it intersects or impinges on a target **216**, such as an input surface or another object. The gap of distance L from the emission point through the reflector **204***a* of the emitted laser light to the target **216** is termed the feedback cavity **208**. The length L of the feedback cavity **208** is variable as the target **216** can move with respect to the laser device.

(40) After the emitted light comes into contact with the target **216**, part or all of the emitted light may be reflected back into the laser cavity **206** as optical feedback. The reflected light enters the laser cavity **206** to coherently or partially coherently interact with the generated laser light. This results in a new state illustrated with a novel emission, P.sub.0-ΔP(L). The new emitted laser light at the new state may have characteristics (e.g., a wavelength or power) that differ from what the emitted laser light would have in the absence of reflection and self-mixing interference. Properties of the novel emission may change depending on a length of the feedback cavity **208**.

(41) The reflector **204***a* may be part of a MEMS structure and may be moveable with respect to reflector **202**, thereby changing a length of the laser cavity **206** and a wavelength of emitted light. The reflector **204***a* may be moveable by a predetermined distance **205**. The predetermined distance **205** may correspond to a value of about 100 nanometers (nm) to 1000 nanometers (nm), though the predetermined distance **205** may be any distance according to principles of the provided disclosure.

(42) FIG. **2**B is a graph **220** showing the variation in power of the novel emission as a function of the length L of the feedback cavity **208**. The graph depicts a predominantly sinusoidal variation with a period of λ/2. Theoretical considerations imply that the variation is given by the proportionality relationship: $\Delta P \propto \cos(4\pi L/\lambda)$. This relationship generally holds in the absence of a strong specular reflection. In the case of such strong specular reflection, the cosine becomes distorted and higher harmonics are present in the relationship. However, the peak-to-peak separation stays at λ/2. For an initially stationary target **216**, this relationship can be used to determine that a deflection has occurred. In conjunction with other techniques, such as counting of the completed number of periods, the range of the deflection may also be determined.

(43) Though the graph **220** shows the variation in power of the new emitted laser light **214** as a

function of the length L of the feedback cavity **208**, similar results and/or graphs may hold for other interferometric properties of a VCSEL or other types of lasers and/or laser diodes that are measured by a self-mixing sensor. Measurements of one or more such interferometric parameters by a self-mixing sensor can be used to infer displacements or motions of the target **216** with respect to a laser.

(44) FIG. **2**C depicts time correlated graphs **230** relating to a laser current **232** (also called a modulation current) with the resulting laser wavelength **234** and the resulting signal **236** of the measured interferometric parameter. The graphs are under the condition of a user input. By driving a laser with a modulation current, such as the laser current **232**, the produced laser light has a laser wavelength **234** that similarly varies according to a triangle wave. As a result of the user input on the touch input surface, the self-mixing interference causes the signal **236** of the interferometric parameter to have the form of a sinusoid (or distorted sinusoid) imposed on a triangle wave. One use of applying the modulation current **232** with a triangle wave is to allow for separate spectrum analyses (e.g., fast Fourier Transforms) of samples taken during the time intervals of the ascending segment and of the descending segment of the triangle waveform modulation of the laser current **232**. While the graphs **230** are shown for a triangle waveform modulation of laser current **232**, some embodiments may use other alternatingly ascending and descending modulation currents for the laser. Also, while the laser current **232** is shown with equal ascending and descending time intervals, in some embodiments these time intervals may have different durations.

(45) FIG. **3** illustrates an example configuration of a self-mixing sensor **300** that may include a top epitaxial stack **302***b* comprising a VCSEL active area and a bottom epitaxial stack **302***a* comprising a photodetector and vertically adjacent to the top epitaxial stack **302***b*, which may be referenced as a VCSEL-integrated photodetector (ViP). In some embodiments, the bottom epitaxial stack **302***a* comprises a resonant cavity photodetector (RCPD). In addition, a microelectromechanical system (MEMS) **302***c* may be vertically adjacent to the top epitaxial stack **302***b*. The self-mixing sensor **300** may be manufactured and epitaxially grown by the process described with respect to FIG. **10** or may be manufactured in accordance with any suitable process. The vertical stack combination of the top epitaxial stack **302***b*, the bottom epitaxial stack **302***a*, and the MEMS **302***c* may provide resonance for VCSEL and RCPD operations.

(46) The semiconductor layers for the bottom epitaxial stack **302***a* may be epitaxially formed on a substrate **306**, such as an n-type substrate. An electrical supply layer **304** may additionally be formed in or on the substrate **306**, such as on a side of the substrate **306** opposite to the bottom epitaxial stack **302***a*, and may be supplied with a reverse biased voltage. In some embodiments, the electrical supply layer **304** is grounded.

(47) The bottom epitaxial stack **302***a* may additionally be structured with n-type distributed Bragg reflectors (nDBR) **308**. The nDBR **308** may include a set of Bragg pairs of alternating materials having different refractive indices. One or more of the materials in the nDBR **308** are doped to be n-type and so form a part of the anode section of a p-n diode structure. An exemplary pair of materials that may be used to form the nDBR **308** are aluminum arsenide (AlAs) and gallium arsenide (GaAs). Though the nDBR **308** is described as an n-type DBR, the nDBR **308** may be doped to be any type. In some cases, the nDBR **308** layers may be formed from reflectors other than distributed Bragg reflectors or from a combination of other reflectors and DBR layers. Though the bottom epitaxial stack **302***a* is depicted as comprising multiple semiconductor layers, in some cases, a bottom epitaxial stack or photodetector may comprise a single layer.

(48) A photodetector active region layer **310** may be formed on top of the nDBR **308**. The photodetector active region layer **310** may be a photon absorption layer, such as an indium gallium arsenide (InGaAs) layer. The photodetector active region layer **310** may, in some embodiments, have a thickness between 15 nm and 1000 nm, though the thickness is not limited to any particular value or range. In some cases, the photodetector active region layer **310** is formed from one or more quantum wells, such as in a single quantum well or multiple quantum well (MQW) structure.

(49) The top epitaxial stack **302***b* may be epitaxially grown on top of the bottom epitaxial stack **302***a*, as depicted in FIG. **3** and as discussed herein. The top epitaxial stack **302***b* may include a VCSEL active region layer **312** that functions, in part, as a lasing layer. The VCSEL active region layer **312** may, in some embodiments, have a thickness between 15 nm and 1000 nm, though the thickness is not limited to any particular value or range. In some cases, the VCSEL active region layer **312** may be formed from one or more quantum wells. An oxide layer **314** may be adjacent to the VCSEL active region layer **312** and may have an aperture through which laser light is guided and through which a current applied to the VCSEL is constricted.
(50) The top epitaxial stack **302***b* may additionally be structured with p-type distributed Bragg reflectors (pDBR) **316**. As discussed in relation to the nDBR **308**, the pDBR **316** may include a set of Bragg pairs of alternating materials having different refractive indexes. Such materials may include AlAs and GaAs. The pDBR **316** may be formed from one or more materials that are doped to be p-type and so form a part of the cathode section of a p-n diode structure. The pDBR **316** layers may be optional and may be omitted in certain embodiments. An electrical contact **324** may be affixed to a topmost layer of the top epitaxial stack **302***b* and may receive a bias current, such as I.sub.BIAS, as depicted in FIG. **1**A, which may induce the emission of laser light **328** from the top epitaxial stack **302***b*. In some cases, the top epitaxial stack **302***b* may operate with a single mode on either an oxide aperture or high-contrast grating. A top surface of the top epitaxial stack **302***b* may additionally be passivated through processes such as atomic layer passivation to prevent or reduce further chemical processes and/or environmental aggressors which may degrade the top epitaxial stack **302***b*.
(51) A MEMS structure **302***c* may be formed on a top surface of the top epitaxial stack **302***b*. The MEMS structure **302***c* may be formed from a sacrificial layer **320** and a partially reflective layer **322**. The sacrificial layer **320** may be used to elevate the partially reflective layer **322** above the VCSEL **302***b* and may be formed of the same material as the pDBR **316** layers or undoped epitaxial layers such as intrinsic Gallium Arsenide (GaAs), though the sacrificial layer **320** may not be used as part of a lasing cavity. The partially reflective layer **322** may be formed as a vertical semiconductor stack including one or more distributed Bragg reflectors. In some cases, the partially reflective layer **322** is a dielectric stack providing partial cavity reflectance and output coupling. Materials for a potential dielectric stack include titanium dioxide (TiO.sub.2) and silicon dioxide (SiO.sub.2).
(52) The partially reflective layer **322** of the MEMS structure **302***c* may include sub-wavelength lateral structures, such as surface relief grating, high-contrast grating, two-dimensional/binary or multi-step diffractive phase masks, meta-surfaces, and so on. The sub-wavelength lateral structures may provide strong cavity reflectivity, such as reflectivity at or about 99%. In cases where sub-wavelength lateral structures are provided on the partially reflective layer **322**, DBR layers of the top epitaxial stack **302***b* and/or the bottom epitaxial stack **302***a* may be reduced or eliminated. This may reduce an overall length of the cavity and/or self-mixing sensor **300** and may improve a quality of a laser light output (e.g., increasing a stability or changing a mode of the laser light output). Sub-wavelength lateral structures may also assist in defining laser polarization, shaping and steering laser light outputs, and so on. The partially reflective layer **322** may additionally be passivated through processes such as atomic layer passivation to prevent or reduce further chemical processes and/or environmental aggressors which may degrade the partially reflective layer **322**.
(53) The MEMS structure **302***c* may be formed from the same materials as the top epitaxial stack **302***b* (e.g., GaAs) or may be formed of additional materials such as silicon, silicon nitride, and silicon dioxide. In cases where the MEMS structure **302***c* is formed from materials intrinsic to the top epitaxial stack **302***b*, the MEMS structure **302***c* may be formed from etching layers of an epitaxially grown semiconductor stack-up. In cases where the MEMS structure **302***c* is formed from extrinsic materials, the MEMS structure **302***c* may be formed independently of the ViP and may be deposited on a top surface of the top epitaxial stack **302***b* after the top epitaxial stack **302***b*

and/or bottom epitaxial stack **302**_a_ is formed. The MEMS structure **302**_c_, and the partially reflective layer **322** in particular, may be moveable with respect to the top epitaxial stack **302**_b_. The movement may occur as a result of a current and/or voltage being applied to the MEMS structure **302**_c_. In some cases, the MEMS structure **302**_c_ may be fully or partially formed of a shape memory alloy, a piezoelectric material, and so on.

(54) As depicted in FIG. **3**, the semiconductor layers for the bottom epitaxial stack **302**_a_ extend laterally beyond the semiconductor layers forming the top epitaxial stack **302**_b_. Such a configuration may be formed by epitaxially forming all the layers included in the self-mixing sensor initially, and then etching part of the layers of the top epitaxial stack **302**_b_. An electrical supply contact **318** may be affixed to topmost layer of the bottom epitaxial stack **302**_a_ at such a position that extends laterally beyond the VCSEL **302**_b_. The electrical supply contact **318** may be connected to a common ground and may act as a cathode of the top epitaxial stack **302**_b_ and/or an anode of the bottom epitaxial stack **302**_a_. In additional or alternative embodiments, the semiconductor layers for the bottom epitaxial stack **302**_a_ do not extend beyond the semiconductor layers of the top epitaxial stack **302**_b_ on one or multiple sides of the self-mixing sensor **300**. In some embodiments, during formation of a particular layer of the common set of semiconductor layers, techniques such as masking and doping are used to alter a doping or chemical composition of a certain layer.

(55) In operation, the top epitaxial stack **302**_b_ is electrically pumped. A forward voltage bias may be applied to the electrical supply contact **324**, which induces a bias current I.sub.BIAS to flow into the top epitaxial stack **302**_b_ and induce emission of laser light **328**, as described previously. For at least part of the time that the top epitaxial stack **302**_b_ is emitting laser light **328**, the bottom epitaxial stack **302**_a_ may be reverse biased by the application of a higher voltage at the electrical supply layer **304** than at the electrical supply contact **318**, to cause the flow of a photocurrent I.sub.PD. In addition to the emitted laser light **328**, some of the laser light produced in the VCSEL active region layer **312** may be directed downward and be absorbed or received in the reverse biased bottom epitaxial stack **302**_a_ and, specifically, by the photodetector active region layer **310**.

(56) If some of the emitted laser light **328** is reflected from an object (not shown) and is received in the top epitaxial stack **302**_b_, self-mixing interference may occur that induces an alteration in the wavelength and/or power of the emitted laser light **328**. Some of such altered laser light may then be received in the bottom epitaxial stack **302**_a_, producing a measurable change in the photocurrent I.sub.PD, or another electrical or interferometric property of the bottom epitaxial stack **302**_a_. Such a measurable change may be used to infer a distance or a motion of the reflecting object. As the phrase is used herein, self-mixing interference refers to the coherent interference between light reflected from an object external to a laser cavity and light present within the laser cavity. A photodetector may detect the output (e.g., self-mixed light) produced in response to a self-mixing operation.

(57) The MEMS structure **302**_c_ may additionally be electrostatically actuated by applying a voltage bias to an electrical supply contact **326**. The voltage bias applied to the electrical supply contact **326** may be independent from the forward voltage bias applied to the electrical supply contact **324** associated with the top epitaxial stack **302**_b_. The voltage bias applied to the electrical supply contact **326** of the MEMS structure **302**_c_ may be varied to change a property of the partially reflective layer **322**. By increasing or decreasing the applied reverse voltage bias, a distance between the partially reflective layer **322** and a top surface of the top epitaxial stack **302**_b_ may increase or decrease, a curvature of the partially reflective layer **322** may change, and so on. In this way, a lasing cavity is changed. In accordance with optical principles, changing a lasing cavity may change properties (e.g., wavelengths or modes) of an emitted laser light. By variations of the partially reflective layer **322**, control of lasing properties of top epitaxial stack **302**_b_ can be achieved.

(58) In various implementations of the provided disclosure, a first current may be supplied to the

top epitaxial stack **302***b*, a second current, distinct from the first current, may be supplied to the MEMS structure **302***c*, and a third current, distinct from the first and second currents, may be supplied to the bottom epitaxial stack **302***a*. In some implementations, distinct voltages may be supplied to the bottom epitaxial stack **302***a*, the top epitaxial stack **302***b*, and the MEMS structure **302***c*.

(59) The MEMS structure **302***c* may allow tuning at a wavelength-scale accuracy. That is, the partially reflective layer **322** may be controlled to produce a laser light with a particular wavelength. Further, power supplied to the top epitaxial stack **302***b* and a temperature of the self-mixing sensor **300** may remain relatively consistent while tuning through the use of the MEMS structure **302***c*. In this way, tuning may be performed linearly. A tuning range of the self-mixing sensor **300** may be about 1%-5% of an initial laser wavelength produced by the VCSEL **302***b*, though, in some embodiments, a tuning range may be less than 1% and/or over 5%, as discussed above. Further, the MEMS structure **302***c* may be tuned at a megahertz modulation speed, though, in some embodiments, a modulation speed may be faster or slower. As a result of the modulation speed, a low duty cycle may be associated with the self-mixing sensor **300**.

(60) In some cases, the bottom epitaxial stack **302***a* may be electrically pumped in a way so as to act as a second VCSEL. That is, the structure depicted in FIG. **3** may, in some embodiments, comprise two VCSEL structures stacked on one another. This may result in a double-pumped VCSEL.

(61) FIG. **4** illustrates example processing circuitry **400** communicatively coupled with one or more electrical supply contacts of a self-mixing sensor, such as the self-mixing sensor **300** as depicted in FIG. **3**. The processing circuitry **400** may include any number of electrical components and may be provided on a printed circuit board or an integrated circuit (IC) coupled with a self-mixing sensor. The processing circuitry **400** may include analog circuits **404** and digital circuits **402**.

(62) Though forward and reverse biases are described as being applied to certain electrical supply contacts/layers, such description is provided merely for explanatory purposes. As understood by a person skilled in the art, any number of voltage and/or current values may be applied to any provided electrical supply contact/layer so as to create a voltage differential used to operate a VCSEL, photodetector, and/or MEMS. Any voltage, current, or combination thereof suitable to operate any of the components described herein may be used in accordance with the provided disclosure.

(63) Though not depicted in FIG. **4**, the electrical supply contact **318** may be connected to a common ground, or a positively or negatively charged object, and may be electrically coupled to both the VCSEL and the photodetector. As the electrical supply contact **318** is electrically coupled to both the VCSEL and the photodetector, the electrical supply contact **318** may act as a cathode for the VCSEL and as an anode for the photodetector.

(64) The electrical supply layer **304**, as described with respect to FIG. **3**, may be electrically coupled to a photodetector. The circuit electrically coupled to the electrical supply layer **304** may be referenced as a self-mixing analysis circuit **406**. The self-mixing analysis circuit **406** may include a transimpedance amplifier **422**, an analog filter AC coupler **420**, one or more amplifiers **418**, an analog-to-digital converter **416**, time-domain signal pre-conditioning circuit **414**, a fast Fourier transform engine **412**, a frequency-domain signal pre-conditioning circuit **410**, and a distance/velocity estimator **408**. Additional, or fewer, elements may be provided in additional or alternative cases.

(65) In accordance with the provided disclosure, a photodetector may output a self-mixing current signal in response to self-mixing operations of a self-mixing sensor (e.g., a VCSEL-integrated photodetector). The self-mixing current signal may be a triangle signal and may require signal processing before being analyzed.

(66) A transimpedance amplifier (TIA) **422** may receive a current produced by the photodetector

and may convert the received current to a voltage output (e.g., acting as a type of current-to-voltage converter). The TIA **422** may be implemented with one or more operational amplifiers. Additionally, the TIA **422** may amplify current signals generated by the photodetector and may minimize instability arising from the junction capacitance of the photodetector. The TIA **422** may have a bandwidth of about 150 kilohertz (kHz) and may have TIA operating noise substantially lower than upstream shot noise.

(67) An analog filter AC coupler **420** may be coupled to an output of the TIA **422** and may suppress DC light intensities from the photodetector. The analog filter AC coupler **420** may also prevent amplifier saturation (e.g., amplifier(s) **418**). The analog filter AC coupler **420** may be a high-pass filter and may have a cut-off frequency of between 10 hertz (Hz) to 100 Hz, or about, in some cases, 40 Hz. In some embodiments, the analog filter AC coupler **420** is omitted. In additional or alternative cases, the analog filter AC coupler **420** may be positioned before the TIA **422**.

(68) One or more amplifiers **418** may additionally be provided as coupled to the TIA **422** and/or the analog filter AC coupler **420**. The one or more amplifiers **418** may be used to boost an amplitude of a signal before processing the signals received by the photodetector, resulting in a positive gain. As a result of the amplified signal, changes in photodetector signals may become more apparent. An analog-to-digital converter (ADC) **416** may be coupled to the one or more amplifiers **418** to convert an analog signal to a digital signal in order to prepare photodetector signals for analysis through sampling and quantizing an input signal. The ADC **416** may have a sampling rate of between 1-5 mega-samples per second (MSPS), may be a 12-bit successive approximation ADC, and may use both shot noise and quantization noise to determine an associated cascaded noise figure of the ADC **416**.

(69) After converting an analog signal to a digital signal, as indicated by the transition between the analog circuits **404** and the digital circuits **402**, processing functions may be performed to determine displacement or motion of an object sensed by the self-mixing sensor, as discussed herein.

(70) The self-mixing analysis circuit **406** may include both time-domain signal preconditioning circuits **414** and frequency-domain signal pre-conditioning circuits **410**, though in some cases one or both of these circuits may be omitted or may be located along a different portion of the self-mixing analysis circuit **406**. The time-domain signal preconditioning circuits **414** may interrogate and/or analyze a self-mixing signal in the time domain and the frequency-domain signal pre-conditioning circuits **410** may interrogate and/or analyze a self-mixing signal in the frequency domain.

(71) A fast Fourier transform (FFT) engine **412** may include a FFT and tone extraction circuit and may apply a received self-mixing signal. From the FFT spectra obtained, a property of the target (e.g., an absolute distance and directional velocity) may be inferred as indicated by distance/velocity estimator **408**. The frequency domain analysis, as indicated by the frequency-domain signal pre-conditioning circuits **410**, may isolate signals corresponding to a change in an operational parameter of a VCSEL and may be used to measure real-world events (e.g., a gesture, a distance between a target and a VCSEL, a speed of the target or the VCSEL, and so on). Values associated with the real-world events may be determined as indicated by the distance/velocity estimator **408**.

(72) The scan scheduler **424** may be any tool, algorithm, and/or circuit used to instruct waveform generators (e.g., laser driving current waveform generator **426** or MEMS voltage waveform generator **428**) which waveform, frequency, and so on to apply to, for example, a VCSEL or a MEMS. In some cases, the scan scheduler **424** instructs one of the laser driving current waveform generator **426** or the MEMS voltage waveform generator **428** to begin or stop applying a waveform to a MEMS or a VCSEL.

(73) The laser driving current waveform generator **426** may generate a current to operate a VCSEL, which may be applied to the electrical contact **324** as depicted in FIG. **3**. The waveform generated

by the laser driving current waveform generator **426** may be a symmetric, triangular waveform and may have a frequency of about 2 kilohertz, though any value may be used in accordance with the provided disclosure. A digital-to-analog converter (DAC) **430***b*, which may have an 8-bit conversion rate of 1 MSPS, may convert the waveform to an analog signal and a current supply **432** may convert a voltage having the waveform generated at the laser driving current waveform generator **426** to a current, thereby driving an associated VCSEL (e.g., via the electrical contact **324**). In some cases, a filter (e.g., a low-pass filter) may additionally be provided to prevent frequency signals higher than a cutoff frequency (e.g., 10 hertz) from being applied to the current supply **432**.

(74) A second output of the laser driving current waveform generator **426** may lead to an additional DAC **430***a* and to the self-mixing analysis circuit **406**. This circuit may act as a waveform correction and may be omitted in some embodiments.

(75) The MEMS voltage waveform generator **428** may generate a driving voltage of a MEMS through, for example, electrical contact **326** as depicted in FIG. **3**. The driving voltage may include a DC and/or AC bias voltage and may cause movement of the associated MEMS (e.g., to increase or decrease a lasing cavity, as discussed above with respect to FIG. **3**). The driving voltage may be a periodic sinusoidal wave, when being applied to the MEMS, and may be terminated when a tuning process is not occurring. The driving voltage may be applied to the electrical contact **326** through a DAC **430***c* and a voltage amplifier **434**.

(76) Though certain circuits and electrical elements are discussed with respect to FIG. **4**, these circuits and elements are provided merely for explanatory purposes. Any number of circuits may be utilized to drive a VCSEL, MEMS, and/or photodetector and to analyze self-mixing signals.

(77) FIGS. **5**A-**5**C depict an example flip-chip self-mixing sensor **500***a*-**500***c*. The self-mixing sensor **500***a*-**500***c* may share similar features with the self-mixing sensor **300** of FIG. **3** and duplicative description may be omitted.

(78) As discussed with respect to FIG. **3**, a MEMS **302***c* may be positioned on a top surface of a top epitaxial stack **302***b*. As depicted in FIG. **5**A, with respect to the self-mixing sensor **500***a*, a MEMS **502***c***1** may be positioned on a bottom surface of an epitaxial stack **502***b*. The epitaxial stack **502***b* may be, or otherwise include, a VCSEL. As the MEMS **502***c***1** is positioned in a cavity formed between the epitaxial stack **502***b* and photodetector **530**, foreign substances may be less likely to come into contact with a surface of the MEMS **502***c***1**.

(79) A semiconductor substrate **528**, such as i-GaAs, may form the top portion of the self-mixing sensor **500***a*. The semiconductor substrate **528** may include an optional on-chip lens formed on a top surface thereof. The semiconductor substrate **528** may shape laser light generated by the epitaxial stack **502***b* and may act as a reflector on one end of a laser cavity. The semiconductor substrate **528** may be curved to focus light to a certain point, or mode, and may shape a beam of light. The semiconductor substrate **528** may be omitted in some embodiments.

(80) The epitaxial stack **502***b* may additionally include a p-type DBR (pDBR) **516** and an n-type DBR (nDBR) **508** which may have alternating materials with different refractive indices. One or more of the materials in the nDBR **508** are doped to be n-type and so form a part of the anode section of a p-n diode structure and one or more of the materials in the pDBR **516** are doped to be p-type and so form a part of the cathode section of a p-n diode structure. An exemplary pair of materials that may be used to form the nDBR **508** and/or the pDBR **516** are aluminum arsenide (AlAs) and gallium arsenide (GaAs). Though the nDBR **508** is described as an n-type DBR and the pDBR **516** is described as a p-type DBR, the nDBR **508** and the pDBR **516** may be doped to be any type. In some cases, the nDBR **508** and the pDBR **516** layers are formed from reflectors other than distributed Bragg reflectors.

(81) The epitaxial stack **502***b* may additionally include a VCSEL active region layer **512** that functions, in part, as a lasing cavity. The VCSEL active region layer **512** may, in some embodiments, have a thickness between 15 nm and 60 nm, though the thickness is not limited to

any particular value or range. In some cases, the VCSEL active region layer **512** may be formed from one or more quantum wells. An oxide layer **514** may be adjacent to the VCSEL active region layer **512** and may have an aperture through which laser light is emitted. A number of electrical contacts **518**, **526***a* may be provided on one of the MEMS **502***c***1** or the epitaxial stack **502***b* to provide a voltage or a current to any of the described components, as discussed with respect to FIGS. **3** and **4**.

(82) The MEMS **502***c***1** may include a partially reflective layer **522***a*, which may be moveable with respect to the lasing cavity of the epitaxial stack **502***b*. As discussed with reference to FIG. **3**, the partially reflective layer **522***a* may move to change a length of the lasing cavity, thereby tuning, or changing a wavelength of, emitted laser light. The MEMS **502***c***1** may additionally include a sacrificial layer **520***a* used to elevate the MEMS **502***c***1** above/below a surface of the epitaxial stack **502***b*. As discussed with respect to FIG. **3**, the MEMS **502***c***1** may be formed from semiconductor layers (e.g., intrinsic GaAs/InP materials) and/or from dielectric materials (e.g., external Si/SiN/SiO.sub.2 materials).

(83) A photodetector **530** may be provided below the MEMS **502***c***1**, such that an object positioned between the photodetector **530** and the MEMS **502***c***1** may be analyzed through self-mixing operations (e.g., through self-mixing signals received at the photodetector **530**). The photodetector **530** may be a silicon-based photodetector.

(84) FIG. **5**B illustrates an example self-mixing sensor **500***b*. Identically numbered components illustrated in FIG. **5**B may be similar to those discussed with respect to FIG. **5**A and duplicative description is omitted for clarity.

(85) The self-mixing sensor **500***b* may have a MEMS **502***c***2** positioned on a top surface of a photodetector **530**. The MEMS **502***c***2** may include a sacrificial layer **520***b*, a partially reflective layer **522***b*, and an electrical contact **526***b* configured to receive a signal that varies a property of the partially reflective layer **522***b*. The MEMS **502***c***2** may be formed on the photodetector **503** and the partially reflective layer **522***b* may be configured (e.g., via an electrical signal received through the electrical contact **526***b*) to move with respect to the photodetector **530**.

(86) FIG. **5**C illustrates an example self-mixing sensor **500***c*. Identically numbered components illustrated in FIG. **5**C may be similar to those discussed with respect to FIGS. **5**A-**5**B and duplicative description is omitted for clarity.

(87) The self-mixing sensor **500***c* may have a MEMS **502***c***3** positioned on a top surface of a nDBR **508**. As illustrated in FIG. 5C, the semiconductor substrate **528** and the pDBR **516** may be replaced by a surface grating **517**. In some implementations, the grating **517** may be a meta-structure. The MEMS **502***c***3** may include a sacrificial layer **520***c*, a partially reflective layer **522***c*, and an electrical contact **526***c* configured to receive a signal that varies a property of the partially reflective layer **522***c*.

(88) FIG. **6** illustrates an example of a high-powered self-mixing sensor **600** having an expanded-cavity. The high-powered self-mixing sensor **600** may be formed from a top epitaxial stack **602***b*, which may include or comprise a VCSEL, vertically adjacent to a bottom epitaxial stack **602***a*, which may include or comprise a photodetector, which may be referenced as a VCSEL-integrated photodetector (ViP). The bottom epitaxial stack **602***a* may be a resonant cavity photodetector (RCPD). In addition, a MEMS **602***c* may be vertically adjacent to the top epitaxial stack **602***b*.

(89) The self-mixing sensor **600** may be manufactured and epitaxially grown by the process described with respect FIG. **10** or may be manufactured in accordance with any suitable process. When compared with the self-mixing sensor **300** of FIG. **3**, a sacrificial layer **620** may be longer than a sacrificial layer **320** in order to extend a distance between a partially reflective layer **622** of a MEMS **602***c* and a top surface of a top epitaxial stack **602***b*. The sacrificial layer **620** may double a length of a laser cavity, though the length may be increased by any multiplier not limited to two. The longer length of the laser cavity may result in a higher-powered laser beam, a narrower free spectral range, and a narrower tunability range (e.g., wavelength tunability).

(90) The semiconductor layers for the bottom epitaxial stack **602***a* may be epitaxially formed on a substrate **606**, such as an n-type substrate. An electrical supply layer **604** may be formed in or on the substrate, such as on a side of the substrate **606** opposite to the bottom epitaxial stack **602***a*, and may be supplied with a reverse biased voltage. In some embodiments, the electrical supply layer **604** is a ground layer.
(91) The bottom epitaxial stack **602***a* may additionally be structured with n-type DBR (nDBR) **608**. The nDBR **608** may include a set of Bragg pairs of alternating materials having different refractive indices. One or more of the materials in the nDBR **608** are doped to be n-type and so form a part of the anode section of a p-n diode structure. An exemplary pair of materials that may be used to form the nDBR **608** are aluminum arsenide (AlAs) and gallium arsenide (GaAs). Though the nDBR **608** is described as an n-type DBR, the nDBR **608** may be doped to be any type. In some cases, the nDBR **608** layers are formed from reflectors other than distributed Bragg reflectors.
(92) A photodetector active region layer **610** may be formed on top of the nDBR **608**. The photodetector active region layer **610** may be a photon absorption layer, such as an indium gallium arsenide (InGaAs) layer. The photodetector active region layer **610** may, in some embodiments, have a thickness between 15 nm and 1000 nm, though the thickness is not limited to any particular value or range. In some cases, the photodetector active region layer **610** is formed from one or more quantum wells.
(93) The top epitaxial stack **602***b* may be epitaxially grown on top of the bottom epitaxial stack **602***a*. The top epitaxial stack **602***b* may include a VCSEL active region layer **612** that functions, in part, as a lasing cavity. The VCSEL active region layer **612** may, in some embodiments, have a thickness between 15 nm and 1000 nm, though the thickness is not limited to any particular value or range. In some cases, the VCSEL active region layer **612** may be formed from one or more quantum wells. An oxide layer **614** may be adjacent to the VCSEL active region layer **612** and may have an aperture through which laser light is emitted.
(94) The top epitaxial stack **602***b* may additionally be structured with p-type DBR (pDBR) **616**. As discussed in relation to the nDBR **608**, the pDBR **616** may include a set of Bragg pairs of alternating materials having different refractive indexes. Such materials may include AlAs and GaAs. The pDBR **616** may be formed from one or more materials that are doped to be p-type and so form a part of the cathode section of a p-n diode structure. The pDBR **616** layers may be optional and may be omitted in certain embodiments. An electrical contact **624** may be affixed to a topmost layer of the top epitaxial stack **602***b* and may be connected to a bias current, which may induce the emission of laser light from the top epitaxial stack **602***b*. In some cases, the top epitaxial stack **602***b* may operate with a single transverse mode on either an oxide aperture or high-contrast gratings.
(95) A MEMS structure **602***c* may be formed on a top surface of the top epitaxial stack **602***b*. The MEMS structure **602***c* may be formed from a sacrificial layer **620** and a partially reflective layer **622**. The sacrificial layer **620** may be used to extend a lasing cavity of the top epitaxial stack **602***b* significantly and may result in a high-powered laser output. In some cases, the sacrificial layer **620** has a length equal to or greater than a length of the pDBR **616** of the top epitaxial stack **602***b*. The sacrificial layer **620** may be formed of the same material as the pDBR **616** layers, though the sacrificial layer **620** may not be used to receive or transmit light. The partially reflective layer **622** may be formed as a vertical stack including one or more distributed Bragg reflectors and/or one or more gratings such as diffraction gratings. In some cases, the partially reflective layer **622** is a dielectric stack providing partial cavity reflectance and output coupling. Materials for a potential dielectric stack include titanium dioxide (TiO.sub.2) and silicon dioxide (SiO.sub.2).
(96) The partially reflective layer **622** of the MEMS structure **602***c* may be formed from sub-wavelength lateral structures, such as surface relief grating, high-contrast grating, two-dimensional/multi-step binary diffractive phase masks, meta-surfaces, and so on. The sub-wavelength lateral structures may provide strong cavity reflectivity, such as reflectivity at or about

99%. In cases where sub-wavelength lateral structures are provided on the partially reflective layer **622**, DBR layers of the top epitaxial stack **602***b* and/or the bottom epitaxial stack **602***a* may be reduced or eliminated. Sub-wavelength lateral structures may also assist in defining laser polarization, shaping and steering laser light outputs, and so on.

(97) The MEMS structure **602***c* may be formed from the same materials as the top epitaxial stack **602***b* (e.g., GaAs) or may be formed of additional materials such as silicon, silicon nitride, and silicon dioxide. In cases where the MEMS structure **602***c* is formed from materials intrinsic to the top epitaxial stack **602***b*, the MEMS structure **602***c* may be formed from etching layers of an epitaxially grown semiconductor stack-up. In cases where the MEMS structure **602***c* is formed from extrinsic materials, the MEMS structure **602***c* may be formed independently of the ViP and may be deposited on a top surface of the top epitaxial stack **602***b* after the top epitaxial stack **602***b* and/or bottom epitaxial stack **602***a* is formed.

(98) As depicted in FIG. **6**, the semiconductor layers for the bottom epitaxial stack **602***a* extend beyond the semiconductor layers forming the top epitaxial stack **602***b*. Such a configuration may be formed by epitaxially forming all the layers included in the self-mixing sensor initially, and then etching part of the layers of the top epitaxial stack **602***b*. An electrical supply contact **618** may be affixed to topmost layer of the bottom epitaxial stack **602***a* at such a position that extends laterally beyond the top epitaxial stack **602***b*. The electrical supply contact **618** may be connected to a common ground and may act as a cathode of the top epitaxial stack **602***b* and/or an anode of the bottom epitaxial stack **602***a*. In additional or alternative embodiments, the semiconductor layers for the bottom epitaxial stack **602***a* do not extend beyond the semiconductor layers of the top epitaxial stack **602***b* on one or multiple sides of the self-mixing sensor **600**. In some embodiments, during formation of a particular layer of the common set of semiconductor layers, techniques such as masking and doping are used to alter a doping or chemical composition of a certain layer.

(99) In operation, the top epitaxial stack **602***b* is electrically pumped. A forward voltage bias is applied to the electrical supply contact **624**, which induces a bias current I.sub.BIAS to flow into the top epitaxial stack **602***b* and induce emission of laser light. For at least part of the time that the top epitaxial stack **602***b* is emitting laser light, the bottom epitaxial stack **602***a* may be reverse biased by the application of a voltage at the electrical supply layer **604**, to cause the flow of a photocurrent I.sub.PD. In addition to the emitted laser light, some of laser light produced in the VCSEL active region layer **612** may be directed downward and be absorbed or received in the reverse biased bottom epitaxial stack **602***a* and, specifically, by the photodetector active region layer **610**.

(100) If some of the emitted laser light is reflected from an object (not shown) and is received in the top epitaxial stack **602***b*, self-mixing interference may occur that induces an alteration in the wavelength of the emitted laser light. Some of such altered laser light may then be received in the bottom epitaxial stack **602***a*, producing a measurable change in the photocurrent I.sub.PD, or another electrical or interferometric property of the bottom epitaxial stack **602***a*. Such a measurable change may be used to infer a distance or a motion of the reflecting object.

(101) The MEMS structure **602***c* may additionally be electrostatically actuated by applying a voltage bias to an electrical supply contact **626**. The voltage bias applied to the electrical supply contact **626** may be independent from the forward voltage bias applied to the electrical supply contact **624** associated with the top epitaxial stack **602***b*. The voltage bias applied to the electrical supply contact **626** of the MEMS structure **602***c* may be varied to change a property of the partially reflective layer **622**. By increasing or decreasing the applied reverse voltage bias, a distance between the partially reflective layer **622** and a top surface of the top epitaxial stack **602***b* may increase or decrease, a curvature of the partially reflective layer **622** may change, and so on. In this way, a lasing cavity is changed. In accordance with optical principles, changing a lasing cavity may change properties (e.g., wavelengths or modes) of an emitted laser light. By variations of the partially reflective layer **622**, control of lasing properties of top epitaxial stack **602***b* can be

achieved. Due to the extended cavity arising from a long sacrificial layer **620**, the self-mixing sensor **600** may have a narrow free spectral range and/or a narrow tunability, with respect to the self-mixing sensor **300** depicted and described with respect to FIG. **3**.

(102) The MEMS structure **602***c* may allow tuning at a wavelength-scale accuracy. That is, the partially reflective layer **622** may be controlled to produce a laser light with a particular wavelength. Further, power supplied to the top epitaxial stack **602***b* and a temperature of the self-mixing sensor **600** may remain relatively consistent while tuning through the use of the MEMS structure **602***c*. In this way, tuning may be performed linearly. A tuning range of the self-mixing sensor **600** may be about 1%-5% of an initial laser wavelength produced by the top epitaxial stack **602***b*, though, in some embodiments, a tuning range may be less than 1% and/or over 5%. Further, the MEMS structure **602***c* may be tuned at a megahertz modulation speed, though, in some embodiments, a modulation speed may be faster or slower. As a result of the modulation speed, a low duty cycle may be associated with the self-mixing sensor **600**.

(103) FIG. **7**A illustrates an example self-mixing sensor array **700***a*. Multiple self-mixing sensors, including the self-mixing sensors described herein, may be coupled together to create an array of sensors, so as to create a number of coherent pixels. In the embodiment depicted in FIG. **7**A, a single, shared MEMS structure **702***c* may be positioned above a first VCSEL portion **702***b***1** and a second VCSEL portion **702***b***2**. Both the first VCSEL portion **702***b***1** and the second VCSEL portion **702***b***2** may be coupled to a shared nDBR **708**, which may extend beneath both the first VCSEL portion **702***b***1** and the second VCSEL portion **702***b***2**. A shared semiconductor substrate **728** may optionally be provided to a surface of the shared nDBR **708** and may reflect internal laser light within a lasing cavity and/or focus emitted laser light, as discussed above.

(104) The first VCSEL portion **702***b***1** may be similar to the second VCSEL portion **702***b***2**. The first VCSEL portion **702***b***1** may be formed from a first VCSEL active region layer **712***a*, a first oxide layer **714***a*, and a first pDBR **716***a*. The functionality of these elements are discussed with respect to other figures, such as FIGS. **5**A-**5**C, above. Likewise, the second VCSEL portion **702***b***2** may be formed from a second VCSEL active region layer **712***b*, a second oxide layer **714***b*, and a second pDBR **716***b*. In some embodiments, the aforementioned layers are common layers epitaxially deposited on the shared nDBR **708** and the layers are etched to form a trench between the first VCSEL portion **702***b***1** and the second VCSEL portion **702***b***2**.

(105) A first electrical contact **724***a* may be provided on the first VCSEL portion **702***b***1** and a second electrical contact **724***b* may be provided on the second VCSEL portion **702***b***2**, so to independently drive the respective VCSEL structures. In some cases, a common electrical contact is provided such that the VCSEL structures are driven identically.

(106) The MEMS structure **702***c* may be formed from a sacrificial layer **720** and a partially reflective layer **722**. The structure and purpose of the sacrificial layer **720** and the partially reflective layer **722** may be similar to those discussed herein and are not repeated here. In the embodiment depicted in FIG. **7**A, the partially reflective layer **722** may extend over both the first VCSEL portion **702***b***1** and the second VCSEL portion **702***b***2**. Therefore, a tuning process (e.g., changing a lasing cavity length) may be performed across both VCSEL structures. The MEMS structure **702***c* may additionally be provided with an electrical contact **726** to provide a driving voltage to the MEMS structure **702***c*, as discussed above. A silicon photodetector, though not depicted in FIG. **7**A, may additionally be placed below the MEMS structure **702***c* to detect self-mixing signals and to transmit the self-mixing signals to processing circuitry.

(107) A photodetector **730** may be provided below the MEMS **702***c*, such that an object positioned between the photodetector **730** and the MEMS **702***c* may be analyzed through self-mixing operations (e.g., through self-mixing signals received at the photodetector **730**). The photodetector **730** may be a silicon-based photodetector and may be positioned on a substrate.

(108) The photodetector **730** may comprise a single photodetector pixels or may comprise an array of photodetectors (e.g., an array of an ‘N’ number of photodetectors). A ratio between a number of

light emitters (e.g., a VCSEL) and a number of light receivers (e.g., a photodetector) may be set to any value. For example, the number of light emitters may be equivalent to the number of light receivers (e.g., a ratio of 1:1). In some embodiments, the number of light emitters may be twice the number of light receivers (e.g., a ratio of 2:1). Other ratios may be used in accordance with the provided disclosure. The light emitters and the light receivers may be individually addressable or may be collectively addressable in any segmentation thereof.

(109) In the embodiment depicted in FIG. **7**A, a single MEMS structure **702***c* is shared across both of the VCSEL pixels. In alternative embodiments, each VCSEL pixel may be associated with an individually tunable MEMS structure.

(110) FIG. **7**B illustrates an example self-mixing sensor array **700***b*. Identically numbered components illustrated in FIG. **7**B may be similar to those discussed with respect to FIG. **7**A and duplicative description is omitted for clarity.

(111) In the self-mixing sensor array **700***b*, the semiconductor substrate **728** and the nDBR **708** may be replaced by a surface grating **717**. In some implementations, the grating **717** may be a meta-structure.

(112) FIG. **8** illustrates an example self-mixing sensor **800** including a backside photodetector **802***a*. As depicted in FIG. **8**, a substrate **806** may be provided. A VCSEL **802***b* may be positioned on one side of the substrate **806** and a photodetector **802***a* may be positioned on the opposite side of the substrate **806**. The photodetector **802***a* may include any number of components as discussed herein, such as a photodetector active layer **810** and n-type/p-type semiconductor layers surrounding the photodetector active layer **810**.

(113) In operation, the VCSEL **802***b* may emit laser light through a top surface and a bottom surface of the VCSEL **802***b*. The light emitted through the top surface may impact an object and may reflect back into the VCSEL **802***b* where a self-mixing operation occurs, as discussed herein. The light emitted through the bottom surface may be received by the photodetector **802***a* and properties of the light may be detected.

(114) The VCSEL **802***b* may operate as other VCSELs described above. The VCSEL **802***b* may include a VCSEL active layer **812**, an oxide layer **814**, and n-type/p-type DBR layers surrounding the VCSEL active layer **812** and the oxide layer **814**.

(115) The MEMS structure **802***c* may be positioned on a top surface of the VCSEL **802***b*. The MEMS structure **802***c* may include a sacrificial layer **820** and a partially reflective layer **822**. The MEMS structure **802***c* may operate as other MEMS structures described herein and duplicative description is omitted with respect to FIG. **8**.

(116) FIG. **9** illustrates an example self-mixing sensor **900** including a top photodetector **902***a*. As depicted in FIG. **9**, a substrate **906** may be provided. A VCSEL **902***b* may be positioned on one side of the substrate **906**,

(117) The VCSEL **902***b* may operate as other VCSELs described above. The VCSEL **902***b* may include a VCSEL active layer **912**, an oxide layer **914**, and n-type/p-type DBR layers surrounding the VCSEL active layer **912** and the oxide layer **914**.

(118) The MEMS structure **902***c* may be positioned on a top surface of the VCSEL **902***b* (e.g., between a VCSEL **902***b* and a substrate cap **950**). The MEMS structure **902***c* may include a sacrificial layer **920** and a partially reflective layer **922**. The MEMS structure **902***c* may operate as other MEMS structures described herein and duplicative description is omitted with respect to FIG. **9**.

(119) A substrate cap **950** may additionally be provided on a top surface of the MEMS structure **902***c*. The substrate cap **950** may be supported on multiple sides and may extend to other self-mixing sensors surrounding the self-mixing sensor **900**. The substrate cap **950** may be at least partially transparent to light emitted by the VCSEL **902***b* so that emitted light may pass through the substrate cap **950** and to the photodetector **902***a* and/or detected object. In some cases, the substrate cap **950** is formed from GaAs.

(120) A photodetector **902***a* may be provided on a top surface of the substrate cap **950**. The photodetector **902***a* may include any number of components as discussed herein, such as a photodetector active layer **910** and n-type/p-type semiconductor layers surrounding the photodetector active layer **910**. Laser light emitted by the VCSEL **902***b* may pass through the MEMS structure **902***c* and the substrate cap **950** and may be received by the photodetector **902***a*, where a measurement of properties of the emitted light is performed. Other self-mixing operations may additionally occur, as discussed above.
(121) FIG. **10** is a flowchart depicting an example manufacturing process **1000** of a ViP including a MEMS structure, according to any of the previously described examples. FIG. **10** is only one example of a potential manufacturing process, and any number of processes could be used to create the products discussed herein.
(122) At operation **1002**, a manufacturing process **1000** begins by epitaxially depositing semiconductor and dielectric layers on a substrate. As discussed above, the substrate may be an n-type substrate and may be formed from Gallium Arsenide (GaAs). Deposited layers may include DBR layers (both n-type and p-type, as discussed above), active laser/photoconductor layers, sacrificial layers for a MEMS, and MEMS layers. The MEMS layers may be formed from a high-contrast reflector dielectric, may be formed from p-type semiconductor layers, and so on.
(123) At operation **1004**, the manufacturing process **1000** may continue by forming one or more polarization setting gratings on a previously formed layer. For example, a polarization setting grating may be formed on a top surface of a partially reflective layer of a MEMS structure to polarize a light output from an associated VCSEL, as discussed above.
(124) At operation **1006**, an etching step may be performed. A partially reflective layer for a MEMS structure may be patterned and/or etched to prepare the remaining, previously epitaxially deposited, layers for future etching or depositing processes. As discussed, the MEMS may be formed from either dielectric layers, semiconductor layers, or any combination thereof and operation **1006** may reference the etching of these layers.
(125) At operation **1008**, a preliminary reactive-ion etching (RIE) process may be performed. The RIE process is a type of dry etching that uses chemically reactive plasma to remove previously deposited material. At operation **1008**, the preliminary RIE may remove certain dielectric and/or semiconductor layers to prepare the semiconductor structure for a subsequent oxidation process.
(126) At operation **1010**, a secondary RIE process may be performed. The secondary RIE process may remove additional material from the semiconductor structure. The secondary RIE process may be performed on different layers, compared with the preliminary RIE process at operation **1008**, and may prepare the semiconductor structure for alignment. The operation at **1008** and **1010** may be performed separately to avoid large amounts of material from being removed at the same time, which may be difficult to efficiently remove effectively. In some cases, the operations at **1008** and **1010** are performed at the same time.
(127) At operation **1012**, certain layers of a semiconductor structure may be oxidized. For example, the RIE processes at **1008** and/or **1010** may uncover portions of certain layers of a semiconductor structure. As some layers are uncovered, an oxidization process may occur with respect to the at least partially uncovered layers. As an example, an oxide layer, as discussed above with respect to, for example, oxide layers **314**/**514**, may be oxidized at this operation.
(128) At operation **1014**, a selective etching (SE) process may be performed. The SE process may selectively etch ends of certain layers, such as an active layer, to prepare the semiconductor structure for the deposition of electrodes and/or for subsequent etching processes. The selective etching process may include a number of discrete etching processes including RIE etching processes, to prepare the semiconductor structure for material deposition steps.
(129) At operation **1016**, electrical conductive material may be deposited at predetermined locations on the semiconductor structure. The electrical conductive material may form electrical contacts, as discussed with respect to, for example, FIGS. **3-5**. The electrical conductive material

may be deposited through any relevant process, including directional evaporation, and may comprise any electrically conductive, or partially electrically conductive, material such as gold.

(130) At operation **1018**, a release step may be performed. The release may be a dry etch release, using a gas such as sulfur hexafluoride. In cases, polymers that form during the release step may be removed by wet cleaning processes. Citric acid may be used in alternative or additional embodiments during the release step.

(131) At operation **1020**, a surface passivation process may be performed. For example, atomic layer deposition (ALD) may be used to coat passivation layers on a surface of a MEMS (e.g., a MEMS grating) and/or VCSEL emitting surface layers. In some implementations, the operation **1020** may be omitted.

(132) The MEMS/ViP structures, as discussed herein, may be used for any number of applications where a self-mixing sensor can be utilized. For example, the MEMS/ViP structures may be provided in an electronic device such as a mobile phone, a smart phone, an electronic watch, a computer, a tablet computer, earbuds or headphones, eyeglasses, and so on. For example, the MEMS/ViP structures may be used as an input device and may detect a user's swipe or gesture. In other examples, the MEMS/ViP structures may detect biological signals from a user, such a heartbeat, blood flow, muscle activity, and so on.

(133) Directional terminology, such as “top”, “bottom”, “upper”, “lower”, “front”, “back”, “over”, “under”, “above”, “below”, “left”, “right”, etc. is used with reference to the orientation of some of the components in some of the figures described below. Because components in various embodiments can be positioned in a number of different orientations, directional terminology is used for purposes of defining relative positions of various structures, and may not always define absolute positions. For example, a first structure described as being “above” a second structure and “below” a third structure is also “between” the second and third structures, and would be “above” the third structure and “below” the second structure if the stack of structures were to be flipped. Also, as used herein, the phrase “at least one of” preceding a series of items, with the term “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase “at least one of” does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes one or more of any of the items, or one or more of any combination of the items, or one or more of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to one or more of only A, only B, or only C; any combination of A, B, or C; and one or more of each of A, B, and C. Similarly, it may be appreciated that an order of elements presented for a conjunctive or disjunctive list provided herein should not be construed as limiting the disclosure to only that order provided.

(134) The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of the specific embodiments described herein are presented for purposes of illustration and description. They are not targeted to be exhaustive or to limit the embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

Claims

1. A self-mixing interferometry (SMI) sensor comprising: a photodetector comprising a first set of semiconductor layers formed on a substrate; a vertical cavity surface emitting laser (VCSEL) stacked with the photodetector and comprising a second set of semiconductor layers formed on the first set of semiconductor layers; and a microelectromechanical system (MEMS) stacked with the VCSEL on a side opposite from the photodetector and comprising an at least partially reflective layer, wherein: the VCSEL is configured to emit light; and the MEMS is configured to change a

length of a resonant cavity associated with the VCSEL, thereby changing a property of the emitted light.
2. The SMI sensor of claim 1, wherein the VCSEL is further configured to: receive reflections of the emitted light after the emitted light interacts with an object; and undergo self-mixing interference after receiving the reflections of the emitted light.
3. The SMI sensor of claim 2, further comprising a processor, wherein: at least a portion of the emitted light is received by the photodetector; and based on at least the portion of the emitted light received by the photodetector, the processor determines at least one of a speed of the object, a roughness of the object, or a distance between the SMI sensor and the object.
4. The SMI sensor of claim 1, wherein the MEMS is separated from the VCSEL by a void in a semiconductor layer.
5. The SMI sensor of claim 1, further comprising processing circuitry configured to: supply a first current to the VCSEL, the first current configured to drive the VCSEL; and supply a second current to the MEMS, the second current separate from the first current and configured to cause a movement of the MEMS.
6. The SMI sensor of claim 1, wherein the changed property of the emitted light is a wavelength of the emitted light.
7. The SMI sensor of claim 6, wherein the at least partially reflective layer of the MEMS is formed from at least one of: a third set of semiconductor layers; or a set of dielectric layers.
8. A tunable sensor for performing self-mixing interferometry, the tunable sensor comprising: a vertical cavity surface emitting laser (VCSEL) comprising a laser cavity; a microelectromechanical system (MEMS) comprising an at least partially reflective layer defining an end of the laser cavity; and a photodetector positioned to receive light emitted by the VCSEL after the light passes through the at least partially reflective layer, wherein a position of the at least partially reflective layer, which is moved when a voltage is applied to the MEMS, defines a length of the laser cavity.
9. The tunable sensor of claim 8, wherein: the VCSEL is a first VCSEL provided on a first set of semiconductor layers; a second VCSEL is provided on the first set of semiconductor layers, the second VCSEL having a second laser cavity and the first VCSEL and the second VCSEL being separated by a trench; and the at least partially reflective layer of the MEMS extends across the first VCSEL and the second VCSEL and further defines a second length of the second laser cavity.
10. The tunable sensor of claim 8, wherein the MEMS changes the length of the laser cavity by moving the at least partially reflective layer from a first position to a second position.
11. The tunable sensor of claim 8, wherein changing the length of the laser cavity results in changing a property of the light emitted by the VCSEL.
12. The tunable sensor of claim 11, wherein the changed property of the light emitted by the VCSEL is a wavelength of the light.
13. The tunable sensor of claim 8, further comprising a condenser lens disposed in a light emission path of the VCSEL on a first surface opposite from a second surface where the MEMS is coupled to the VCSEL.
14. The tunable sensor of claim 13, wherein the VCSEL comprises: an active layer; an oxide layer adjacent to the active layer; a first set of distributed Bragg reflectors positioned between the MEMS and the oxide layer; and a second set of distributed Bragg reflectors positioned between the condenser lens and the active layer.
15. The tunable sensor of claim 8, wherein the photodetector is spaced apart from the MEMS and the VCSEL.
16. A self-mixing sensor comprising: a substrate; a first set of semiconductor layers positioned above the substrate, the first set of semiconductor layers operating as a first light receiving or light emitting element; a second set of semiconductor layers stacked with the first set of semiconductor layers, the second set of semiconductor layers operating as a second light receiving or light emitting element; and a microelectromechanical system (MEMS) stacked with the second set of

semiconductor layers, the MEMS comprising a moveable reflective layer.
17. The self-mixing sensor of claim 16, further comprising processing electronics electrically coupled with the first set of semiconductor layers, the processing electronics configured to analyze electrical current produced by the first set of semiconductor layers to determine self-mixing properties of the light.
18. The self-mixing sensor of claim 16, wherein the first set of semiconductor layers and the second set of semiconductor layers each comprise distributed Bragg reflectors.
19. The self-mixing sensor of claim 16, wherein the moveable reflective layer comprises at least one of a diffractive grating or a subwavelength grating to polarize the light produced by a VCSEL.
20. The self-mixing sensor of claim 16, further comprising a void positioned between the MEMS and the second set of semiconductor layers, the void and the second set of semiconductor layers at least partially defining a laser cavity.